Dual-Task Interactive Learning for Unsupervised Spatio-Temporal–Spectral Fusion of Remote Sensing Images

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***Abstract*— Spatio-temporal–spectral fusion aims to produce high spatio-temporal–spectral resolution images by integrating the complementary spatial, temporal, and spectral advantages of multisource remote sensing images. However, on one hand, the existing spatio-temporal–spectral fusion methods are insufficient to exploit the inherent complex nonlinear spatial, temporal, and spectral relationship among multisource and multitemporal observations. On the other hand, since the unavailability of real high spatio-temporal–spectral resolution images, it is difficult to adopt deep learning (DL) methods with supervised training. In this article, we propose an effective unsupervised spatio-temporal–spectral fusion model (USTSFM) with dual-task interactive learning to alleviate these problems. The proposed USTSFM has two branches: the spatio-temporal–spectral mapping (STSM) branch is to describe the temporal relationship, and the spectral super-resolution (SSR) branch is to model the spectral relationship. Moreover, the spatial–spectral interaction compensation block is designed to make the two branches compensate and benefit from each other. This intrinsically related and mutually facilitated strategy allows the USTSFM to sufficiently exploit the inherent spatial, temporal, and spectral relationship. In addition, a shared reconstruction module is meticulously designed for the two tasks, which not only reduces the parameters but also allows the supervised task to guide the convergence of the unsupervised task, boosting the stability of unsupervised training. The qualitative and quantitative results demonstrated that the proposed USTSFM has richer spatial details and more accurate predictions than the other state-of-the-art methods.**

***Index Terms*— Dual-task interactive learning, remote sensing, spatio-temporal–spectral fusion, unsupervised.**

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I. Introduction

REMOTE sensing images with high temporal, spatial, and spectral resolutions have considerable potential for investigating dynamic monitoring applications, such as environmental monitoring [1], [2], agriculture monitoring [3], and urban change mapping [4]. However, current satellite systems always have a tradeoff among spatial, temporal, and spectral resolutions due to the limitations of hardware and other factors, which limits the development of these applications to a certain extent. For example, spaceborne hyperspectral (HS) imagery is featured with high spectral resolution; however, they are generally characterized by low spatial and temporal resolutions. As shown in Table I, the Hyperion sensor on EO-1 launched in 2000 captured HS images with 242 spectral bands, but the revisit period is 200 days and the spatial resolution is 30 m. The Chinese ZiYuan (ZY)-1 02D satellite launched in 2019 provides HS images with a 55-day revisit period and a spatial resolution of 30 m. On the contrary, satellite multispectral (MS) sensors generally provide images with a higher spatio-temporal resolution, and the spectral resolution is generally low, such as the Sentinel-2 MS imagery. Fortunately, multisource remote sensing spatio-temporal–spectral fusion aims to generate the high spatio-temporal–spectral resolution images by merging the complementary spatial, temporal, and spectral advantages, which is a cost-effective way to alleviate the tradeoff among spatial, temporal, and spectral resolutions of current satellite systems.

To date, a variety of multisource image fusion methods have been proposed [5], [6], [7], which can be broadly divided into four categories: 1) multiview spatial fusion; 2) spatio-spectral fusion; 3) spatio-temporal fusion; and 4) spatio-temporal–spectral fusion.

A. Multiview Spatial Fusion

Multiview spatial fusion tries to generate a higher spatial resolution image by integrating a series of multiview relatively lower spatial resolution images captured by the same imaging system [8], [9]. Multiview spatial fusion can generally be divided into multitemporal and multiangle spatial fusion. Among them, the development of the multitemporal fusion

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TABLE I
SPATIAL, SPECTRAL, AND TEMPORAL RESOLUTIONS OF SEVERAL TYPICAL SATELLITE SENSORS

Sensors	Spatial resolution	Spectral bands	temporal resolution
Hyperion(HS)	30m	242 bands	200 days
ZY-1 02D(HS)	30m	166 bands	55 days
Sentinel-2(MS)	10m	13 bands	10 days
Landsat ETM+ (MS)	30m	8 bands	16 days

method is relatively early; Tsai [10] first introduced multiview spatial fusion based on simulated multitemporal images, which has been used in remote sensing sensors for improving hardware performance. Subsequently, Shen et al. [11] and Merino and Nunez [12] proposed the multitemporal learning fusion methods with the idea of super-resolution for multitemporal images. However, it is difficult to obtain effective fusion results when large changes happen in land cover types. Multiangle remote sensing imaging sensors can quickly obtain observation images of the same area from different angles, which is more suitable for multiview fusion. Therefore, the multiangle fusion method has been developed rapidly, for example, Zhang et al. [13] and Ma et al. [14] further extended the super-resolution analysis to the multiview spatial fusion. However, multiview fusion only focuses on the improvement of spatial resolution and does not consider the spectral resolution, which limits further development.

B. Spatio-Spectral Fusion

Spatio-spectral fusion seeks to generate high spatio-spectral resolution images by complementary integrating the spatial and spectral advantages of multisource remote sensing images [15]. The classic spatio-spectral fusion methods include panchromatic (PAN)/MS fusion [16], [17], PAN/HS fusion [18], [19], and MS/HS fusion [20], [21].

PAN/MS fusion is a popular case of the spatial–spectral fusion method, to generate a relatively higher spatial resolution image by merging a high spatial resolution PAN image with a lower spatial resolution MS image [16]. Currently, the existing PAN/MS fusion methods can be generally divided into four categories [22]: 1) component substitution (CS)-based methods; 2) multiresolution analysis (MRA)-based methods; 3) variational optimization (VO)-based methods; and 4) deep learning (DL)-based methods. Although CS-and MRA-based methods are simple and efficient, they generally introduce spectral distortions or spatial blurring to the fused images [23], [24]. The VO-based methods combine the regularization priors and optimization methods [25], [26], which achieve better fusion performance than the CS-and MRA-based methods. However, it is unsuitable for some real-time applications due to the tradeoff between the fusion performance and the computational cost. Recently, DL has become a hotspot of current research due to its excellent performance, e.g., the pansharpening based on convolutional neural network (PNN) [27], the progressive network with deep spectral transformation for pansharpening [28], and the detail injection model-based deep fusion network (DIM-FuNet) [29]. Moreover, the framework of the transformer [30] was also used for pansharpening, e.g., vision transformer for pansharpening [17] and convolution-embedded vision transformer for pansharpening [31]. The DL-based fusion methods mostly use Wald's protocol [32] to construct training data for supervised learning. Currently, several unsupervised methods have also been proposed to directly train the fusion model on the initial data to alleviate the loss of spatial–spectral information, e.g., the pansharpening based on generative adversarial network (Pan-GAN) [6] and the supervised–unsupervised combined fusion network (SUFNet) [33].

In addition, PAN/HS fusion and MS/HS fusion are the other two crucial spatio-spectral fusion categories. Currently, a variety of PAN/HS fusion and MS/HS fusion methods have been proposed, several of which are extended from the PAN/MS fusion methods [34], [35]. Moreover, the DL-based PAN/HS fusion and MS/HS fusion methods are also springing up due to the amazing performance, e.g., the multiobjective guided divide-and-conquer network for PAN/HS fusion (MO-DCN) [36] and the multiscale low-rank deep back projection fusion network for MS/HS fusion (MLR-DBPFN) [37].

C. Spatio-Temporal Fusion

Spatio-temporal fusion is to generate high spatio-temporal resolution images by aggregating the high spatial but low temporal images and the low spatial but high temporal images. To date, spatio-temporal fusion technology has been developed rapidly, which can be approximately divided into four categories: 1) weighting-based methods; 2) unmixing-based methods; 3) model-based methods; and 4) DL-based methods. The weight-function-based methods and the unmixing-based methods perform well in phenological changes, but it is difficult to obtain a high-fidelity fusion result in the scene with complex land cover types, e.g., the spatial and temporal adaptive reflectance fusion model (STARFM) [38] and the spatio-temporal data fusion model (STDFM) [39]. The model-based methods generally perform better than the weighting-based methods and the unmixing-based methods, such as the error-bound-regularized sparse coding (EBSPTM) [40], and the sparse-representation-based spatio-temporal reflectance fusion model (SPSTFM) [41]. However, these methods have limited performance in land cover changes. Recently, the DL-based methods have gradually been used in spatio-temporal fusion due to their impressive performance. The spatio-temporal fusion with the two-stream convolutional neural network (StfNet) [42] is a classic DL-based spatio-temporal fusion method, which combines temporal consistency and temporal dependence to predict the changes more accurately. Afterward, the DL-based spatio-temporal fusion methods developed rapidly, and a variety of DL-based methods were proposed. For example, the progressive spatio-temporal attention fusion method with the GAN (PSTAF-GAN) [43] and the spatio-temporal fusion using recurrent neural network (STFRNN) [44] can also obtain competitive fusion results. In addition, the framework of the transformer [30] was also used for spatio-temporal fusion due to the ability to capture long-distance features, e.g., the multistage fusion with texture

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transformer and convolutional neural network (CNN) (MSFusion) [45] and the spatio-temporal fusion model using swin transformer (SwinSTFM) [46].

D. Spatio-Temporal–Spectral Fusion

Spatio-temporal–spectral fusion seeks to produce high spatio-temporal–spectral resolution images with complementary integrating the spatial, temporal, and spectral advantages of multisource remote sensing images. Currently, only a few studies focus on exploring the relationship among spatial, temporal, and spectral information. To the best of our knowledge, Shen [47] first conducted research on the integration framework for spatio-temporal–spectral fusion but testing on the simulated data. Subsequently, Shen et al. [48] further proposed an integrated spatio-temporal–spectral fusion framework based on the variational theory and testing on the actual data. Wu et al. [49] and Meng et al. [50] also presented the integrated fusion frameworks, but did not aggregate the spatial, temporal, and spectral features simultaneously. In recent years, Zhao et al. [51] proposed an integrated spatio-spectral–temporal fusion method based on a sparse representation model, which can achieve multiple remote sensing image fusion tasks. Peng et al. [52] proposed an integrated spatio-spectral–temporal fusion framework based on semicoupled sparse tensor factorization. Zhou et al. [53] introduced a spatio-spectral–temporal fusion model based on generalized linear spectral mixing (GLMM-STSF), which is first designed for the characteristics of current HS image sensors. However, the GLMM-STSF insufficiently exploits the complex nonlinear relationship among the spatial, temporal, and spectral due to the linear assumption. Tang et al. [54] presented a spatio-temporal–spectral fusion framework for downscaling Sentinel-3 ocean and land colour imager (OLCI), which consists of spatio-temporal fusion and spatio-spectral fusion.

E. Challenges and Contributions

Although few spatio-temporal–spectral fusion methods have been proposed, some major challenges still exist, *Challenge 1:* Most existing spatio-temporal–spectral fusion methods fail to adapt to the characteristics of current satellite sensors, which further limits the development of Earth observation applications. Specifically, the existing spatio-temporal–spectral fusion methods [48], [52] mainly focus on HS images with high temporal–spectral resolutions but low spatial resolution and MS images with high spatial resolution but the low temporal–spectral resolution (e.g., moderate-resolution imaging spectroradiometer (MODIS)-Landsat image pairs) as the research objects. However, in real-world scenarios, the images captured by spaceborne HS satellite sensors, such as Hyperion and ZY-1 02D, usually have low spatial and temporal resolutions. *Challenge 2:* The relationship among multisource spatio-temporal–spectral images is more complicated, and it is difficult to synergistically represent the complex nonlinear relationship among the spatial, temporal, and spectral information. *Challenge 3:* Although DL has a strong nonlinear presentation ability, there is no real high spatio-temporal–spectral resolution image as a reference for supervised training.

In this article, considering these challenges described above, we propose an effective unsupervised spatio-temporal–spectral fusion model (USTSFM). Generally, we derive the following contributions.

1) To alleviate *Challenge 1*, we propose an unsupervised deep spatio-temporal–spectral fusion method via DL to match the characteristics of current satellite sensors. To the best of our knowledge, this is the first unsupervised deep-learning-based spatio-temporal–spectral fusion model.

2) To overcome *Challenge 2*, we describe the complex relationships among the spatial, temporal, and spectral information in a dual-task interactive learning manner, in which one task for the spatio-temporal–spectral mapping (STSM) is to describe the temporal relationship and the other task for spectral super-resolution (SSR) is to model the spectral relationship. The spatial–spectral interactive compensation block (SSICB) is proposed to represent the spatial relationship and exchange the spatial and spectral information in the two tasks (e.g., STSM and SSR) for compensating for each other. In this way, the relationships among spatial, temporal, and spectral are collaboratively described in a unified architecture.

3) To overcome *Challenge 3*, a shared reconstruction module is meticulously designed for the two tasks. In this way, the supervised STSM task can guide the unsupervised SSR task to converge, which can boost the fusion accuracy of the unsupervised network to generate the high-fidelity fusion image.

The rest of the article is organized as follows. Section II introduces the proposed fusion network. Section III presents the experimental results, and finally, the conclusion is given in Section IV.

II. Proposed Fusion Network

A. Network Architecture

In this article, we use lowercase and bold letters to denote scalars and matrices, respectively. Specifically, let $\mathbf{Y}_i \in R^{H\times W\times b}$ represent the MS image at date i, $\mathbf{X}_j \in R^{h\times w\times B}$ denote the HS image at date j, $H\times W$ and $h\times w$ are the size of the spatial dimension of MS and HS images, respectively, and b and B are the number of spectral bands of MS and HS images, respectively. In general, image $\mathbf{X}$ has a higher spectral resolution and lower spatial–temporal resolution, and image $\mathbf{Y}$ has a higher spatial–temporal resolution and a lower spectral resolution. To completely describe the complex nonlinear relationship among the spatial, temporal, and spectral information, we propose an interactive learning network to construct a bridge between the STSM task and the SSR task in a dual-task framework. Specifically, the STSM task aims to describe the spatial–spectral changes between $\mathbf{X}_i$ and $\mathbf{X}_j$, which is expressed as (1). The SSR task aims to reconstruct the target image $\mathbf{Z}_j$ by super-resolution in the spectral dimension, which is expressed as (2)

$$\mathbf{X}_j = f_{\text{stsm}}((\mathbf{X}_i, \mathbf{Y}_i, \mathbf{Y}_j), \Theta_{f_{\text{stsm}}}) \tag{1}$$

$$\mathbf{Z}_j = f_{\text{SSR}}((\mathbf{Y}_j), \Theta_{f_{\text{SSR}}}) \tag{2}$$

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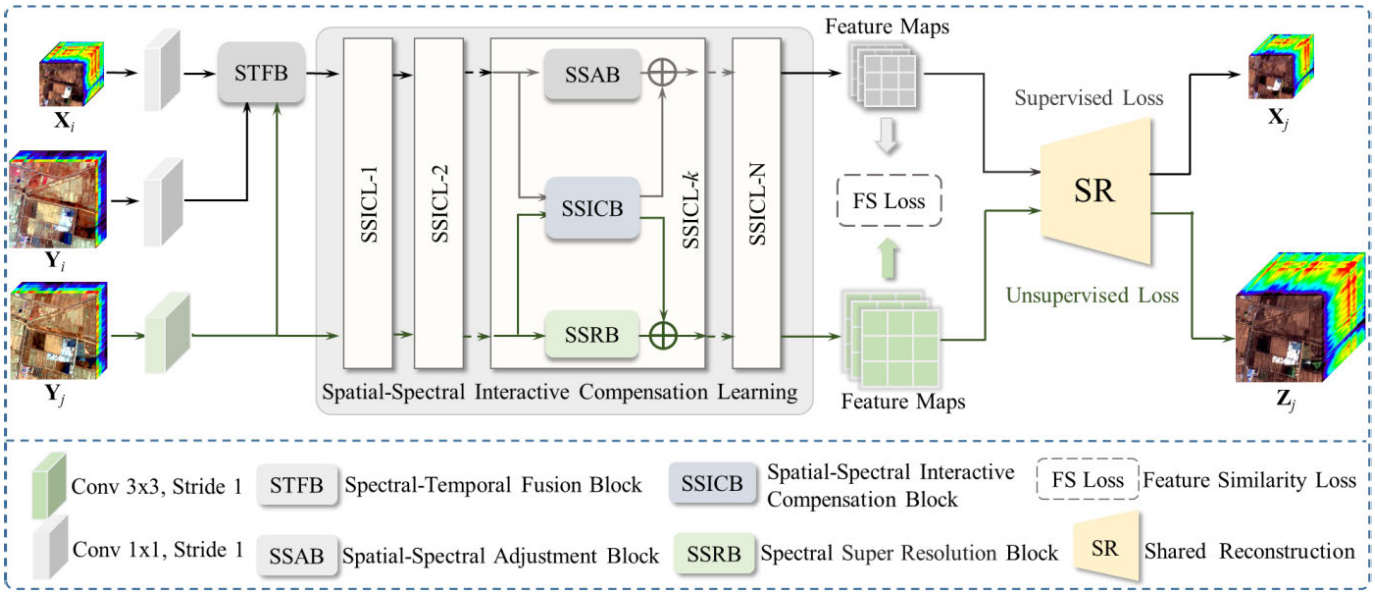


Fig. 1. Overall framework of the proposed USTSFM. $\mathbf{X}$ and $\mathbf{Y}$ are the HS image and MS image, respectively. $\mathbf{Z}$ is the desired fusion HS image, and i and j are the neighboring date and target date of images' acquisition, respectively.

where f_{stsm} is the STSM model, $\Theta_{f_{\text{stsm}}}$ represents the learnable parameters of f_{stsm}, f_{ssr} is the SSR model, and $\Theta_{f_{\text{SSR}}}$ denotes the learnable parameters of f_{ssr}.

The proposed USTSFM takes an HS image before the target date (i.e., $\mathbf{X}_i$), an MS image before the target date (i.e., $\mathbf{Y}_i$), and MS images at the target date (i.e., $\mathbf{Y}_j$) as inputs to generate two images: the HS image at the target date (i.e., $\mathbf{X}_j$) and the target high spatial resolution HS image (i.e., $\mathbf{Z}_j$). The overall framework of USTSFM is illustrated in Fig. 1, which consists of two branches: the STSM network and the SSR network. The STSM network obtains the lower spatial resolution HS images, and the SSR network obtains the higher spatial resolution HS images. Since the goal of both the STSM and SSR networks are to obtain the HS images with different spatial resolutions at the same date. The STSM task can provide the spectral information required by the SSR task, while the SSR task can provide the STSM task with accurate land cover details. Thus, we design a spatial–spectral interactive compensation learning (SSICL) block to make the two networks be compensated for each other. This intrinsically related and mutually facilitated strategy allows the STSM and SSR tasks to benefit from each other and effectively exploit the correlation of spatial, temporal, and spectral to improve the learning ability of the proposed USTSFM. Furthermore, a shared reconstruction module is designed to make supervised learning guide the convergence of unsupervised learning.

1) STSM Network Branch: The STSM network aims to reconstruct the HS image by predicting the spatial–spectral changes. We first use two convolutional layers with the size of 1×1 and 3×3 to extract the spatial–spectral information of HS image and MS image, respectively. Then, we design a spectral–temporal fusion block (STFB) to rectify the spatial–spectral information, which is illustrated in Fig. 2(a). The difference image between MS images (i.e., $\mathbf{Y}_i$ and $\mathbf{Y}_j$) is obtained via subtraction, and then integrating it with the HS image before the target date (i.e., $\mathbf{X}_i$) to obtain the difference image (i.e., $\mathbf{X}_{ij}$) between the HS images. Finally, the preliminary target HS image is obtained by adding the difference image (i.e., $\mathbf{X}_{ij}$) and the HS image before the target date (i.e., $\mathbf{X}_i$). After STFB, the spatial–spectral features are similar to the HS image at the target date, but there are still some gaps. Thus, we design N spatial–spectral adjustment block (SSAB) to correct the spatial–spectral features, which is shown in Fig. 2(b). In the SSAB, spatial attention and spectral attention are, respectively, used to correct spatial–spectral information. Finally, a reconstruction module is designed to reconstruct the HS image at the target date.

2) SSR Network Branch: The SSR network aims to reconstruct the high spatial resolution HS image by taking a super-resolution operation on the MS image in spectral dimension. The designed SSR network takes the MS image at the target date (i.e., $\mathbf{Y}_i$) as input to generate the target high spatial resolution HS image. Similar to the STSM network branch, a convolutional layer with kernel size 3×3 is first leveraged to extract the spatial–spectral features of the MS image. Since the SSR task needs to simultaneously recover both the spatial-structure features and spectral-specific features, we design N SSR block (SSRB) to refine the spatial–spectral information. The overall framework of the SSRB is shown in Fig. 3(a). Specifically, the depthwise convolutional operations for each channel are used to learn the specific spatial–spectral features of each channel, and then the features of each channel are merged. After that, spectral attention [as shown in Fig. 3(b)] and spatial attention [as shown in Fig. 3(c)] are used to adaptively learn the valuable spatial–spectral features. Then a pointwise convolutional layer is used to integrate the spatial–spectral information. Finally, a reconstruction module is designed to reconstruct the target high spatial resolution HS image.

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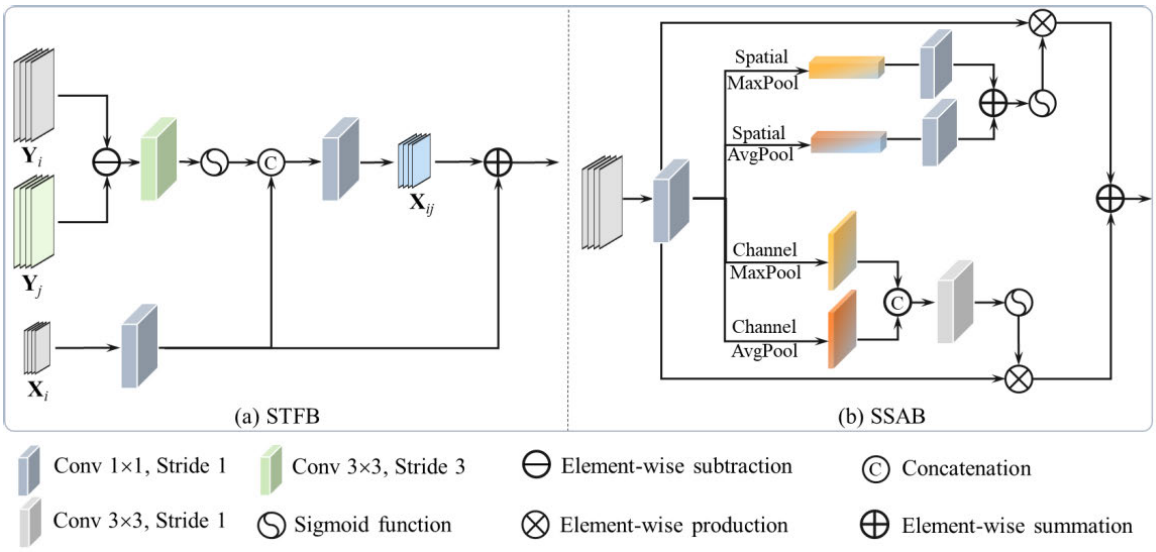


Fig. 2. Framework of STFB and SSAB. $\mathbf{X}_{ij}$ is the difference image learned from $\mathbf{Y}_i$, $\mathbf{Y}_j$, and $\mathbf{X}_i$. (a) STFB. (b) SSAB.

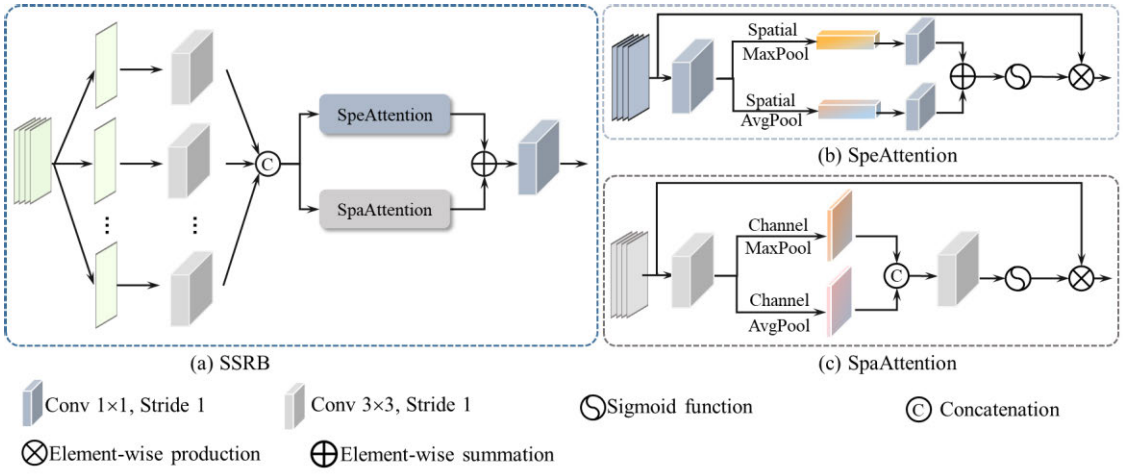


Fig. 3. Framework of SSRB. (a) SSRB. (b) Spectral attention (SpeAttention). (c) Spatial attention (SpaAttention).

3) Spatial–Spectral Interactive Compensation Block (SSICB): The STSM network branch has a wealth of spectral information, which is required by the SSR branch, and the SSR network branch contains accurate spatial information and partial spectral information, which is required by the STSM network branch. Therefore, SSICB is designed to allow the STSM network branch and the SSR network branch to exchange the spatial–spectral information and compensate for each other. The framework of SSICB is illustrated in Fig. 4. In the k th SSICB, the inputs (i.e., F^k_{stsm} and F^k_{SSR}) are decomposed into the spatial components (i.e., $F^k_{\text{STSM}_{\text{spa}}}$ and $F^k_{\text{SSR}_{\text{spa}}}$) and spectral components (i.e., $F^k_{\text{STSM}_{\text{spe}}}$ and $F^k_{\text{SSR}_{\text{spe}}}$), respectively. Since the MS images contain rich spatial information and part of spectral information, we use the spatial and spectral components of the SSR task (i.e., $F^k_{\text{SSR}_{\text{spa}}}$ and $F^k_{\text{SSR}_{\text{spe}}}$) to the STSM task for spatial–spectral compensation. After integration, the overall features of the STSM task are transferred to the SSR task to compensate for the spectral information. In this way, the spatial and spectral features are compensated for each other, and finally the spatial components and spectral components are fused to obtain the fusion features. Specifically, the output of the STSM branch in the k th SSICB (i.e., F^{k+1}_{stsm}) is obtained as

$$F^{k+1}_{\text{stsm}} = F^{k+1}_{\text{STSM}_{\text{spa}}} + F^{k+1}_{\text{STSM}_{\text{spe}}} \tag{3}$$

$$F^{k+1}_{\text{STSM}_{\text{spa}}} = \text{SpaAtt} \times \left(\text{Relu}\left(\text{Conv3}\left(\text{Concat}\left(F^k_{\text{STSM}_{\text{spa}}}, DF^k_{\text{SSR}_{\text{spa}}}\right)\right)\right)\right) \tag{4}$$

$$F^{k+1}_{\text{STSM}_{\text{spe}}} = \text{SpeAtt} \times \left(\text{Relu}\left(\text{Conv1}\left(\text{Concat}\left(F^k_{\text{STSM}_{\text{spe}}}, DF^k_{\text{SSR}_{\text{spe}}}\right)\right)\right)\right) \tag{5}$$

$$F^k_{\text{STSM}_{\text{spa}}} = \text{Relu}\left(\text{Conv3}\left(F^k_{\text{stsm}}\right)\right) \tag{6}$$

$$F^k_{\text{STSM}_{\text{spe}}} = \text{Relu}\left(\text{Conv1}\left(F^k_{\text{stsm}}\right)\right) \tag{7}$$

$$DF^k_{\text{SSR}_{\text{spa}}} = \text{Relu}\left(\text{DConv3}\left(\text{Relu}\left(\text{Conv3}\left(F^k_{\text{SSR}}\right)\right)\right)\right) \tag{8}$$

$$DF^k_{\text{SSR}_{\text{spe}}} = \text{Relu}\left(\text{DConv3}\left(\text{Relu}\left(\text{Conv1}\left(F^k_{\text{SSR}}\right)\right)\right)\right) \tag{9}$$

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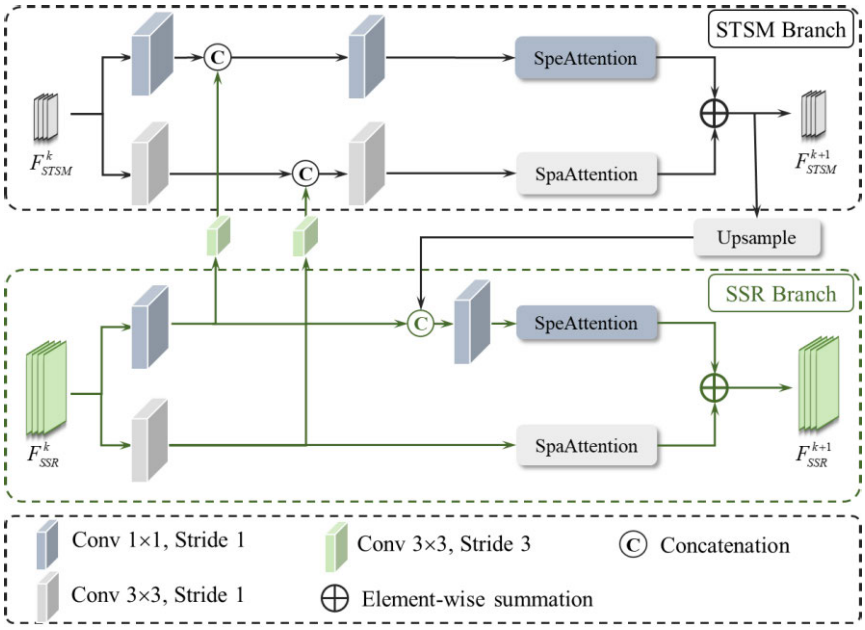


Fig. 4. Framework of SSICB. F^k_{stsm} and F^k_{SSR} are the features of STSM and SSR branches, respectively. F^{k+1}_{stsm} and F^{k+1}_{SSR} are the features of STSM and SSR branches through an SSICB, respectively.

where Conv3 and the Conv1 are the convolution operator with the kernel size of 3×3 and 1×1, respectively, Concat is the concatenation operation, Relu is the rectified linear unit, SpeAtt and SpaAtt denote the spectral attention [as shown in Fig. 3(b)] and spatial attention [as shown in Fig. 3(c)], respectively, and DConv3 is the convolution operator with the kernel size 3×3 and a stride of 3 to make the features have the same spatial size. The output of the SSR branch in the k th SSICB (i.e., F^{k+1}_{SSR}) is obtained as

$$F^{k+1}_{\text{SSR}} = F^{k+1}_{\text{SSR}_{\text{spa}}} + F^{k+1}_{\text{SSR}_{\text{spe}}} \tag{10}$$

$$F^{k+1}_{\text{SSR}_{\text{spa}}} = \text{SpaAtt}\big(\text{Relu}\big(\text{Conv3}\big(F^k_{\text{SSR}}\big)\big)\big) \tag{11}$$

$$F^{k+1}_{\text{SSR}_{\text{spe}}} = \text{SpeAtt}\big(\text{Relu}\big(\text{Conv1}\big(\text{Concat}\big(\text{Up}\big(F^{k+1}_{\text{stsm}}\big), F^k_{\text{SSR}_{\text{spe}}}\big)\big)\big)\big) \tag{12}$$

$$F^k_{\text{SSR}_{\text{spe}}} = \text{Relu}\big(\text{Conv1}\big(F^k_{\text{SSR}}\big)\big) \tag{13}$$

where Up() is the upsampled operation with the bicubic interpolation algorithm.

4) Shared Reconstruction Module: Since the STSM task has spectral high-fidelity capability in HS image reconstruction due to the supervised learning, we use the STSM network as a regularization branch to reconstruct the target high spatial resolution HS image. In addition, the STSM network can assist us to describe the complex nonlinear relationship among spatial, temporal, and spectral. Therefore, we propose a novel shared reconstruction architecture for HS images considering the spatial structure information and spectral detail information, and the framework of the proposed reconstruction is shown in Fig. 5. In this shared reconstruction fashion, the supervised reconstruction of low spatial resolution HS image can assist the unsupervised reconstruction of high spatial resolution HS image to obtain a high-fidelity fusion image. Specifically, we first fed the features F_f into a convolutional layer to obtain the global spectral feature F_g. Then we split the features F_f into four groups in the channel dimension by considering the spectral information in the HS image, which can be expressed as $[F^1_f, F^2_f, F^3_f, F^4_f]$. Moreover, we use dilation convolution to capture the spatial information of HS images using different receptive fields for each group. In addition, the gated recurrent unit (GRU) [55] is used to describe the spatial–spectral correlation information between different groups. Finally, the discriminative features $[F^1_r, F^2_r, F^3_r, F^4_r]$ can be obtained, which can be described as follows:

$$F^1_r = \text{GRU}\big(F_g, \text{Relu}\big(\text{Conv1}\big(F^1_f\big)\big)\big) \tag{14}$$

$$F^2_r = \text{GRU}\big(F^1_r, \text{Relu}\big(\text{DilatedConv1}\big(F^2_f\big)\big)\big) \tag{15}$$

$$F^3_r = \text{GRU}\big(F^2_r, \text{Relu}\big(\text{DilatedConv2}\big(F^3_f\big)\big)\big) \tag{16}$$

$$F^4_r = \text{GRU}\big(F^3_r, \text{Relu}\big(\text{DilatedConv3}\big(F^4_f\big)\big)\big) \tag{17}$$

$$F_g = \text{Relu}\big(\text{Conv1}\big(F_f\big)\big) \tag{18}$$

where GRU() is the gated recurrent unit (GRU), and DilatedConv K is the dilated convolution with the dilation rate K to capture the multiscale features. In this way, the global spectral information F_g can be used as guidance information to learn the relationship between the local spectral information in different groups. Then the reconstructed image is obtained as

$$Z = \text{SpeAtt}(\text{Relu}(\text{Conv1}(F_r))) + \text{SpaAtt}(\text{Relu}(\text{Conv3}(F_r))) \tag{19}$$

$$F_r = \text{Concat}\big(F^1_r, F^2_r, F^3_r, F^4_r\big). \tag{20}$$

B. Loss Function

In this article, to improve the stability of the model, we use the STSM network as a regularization branch, and we use supervised loss and unsupervised loss to jointly optimize our proposed USTSFM. The overall loss function of the USTSFM

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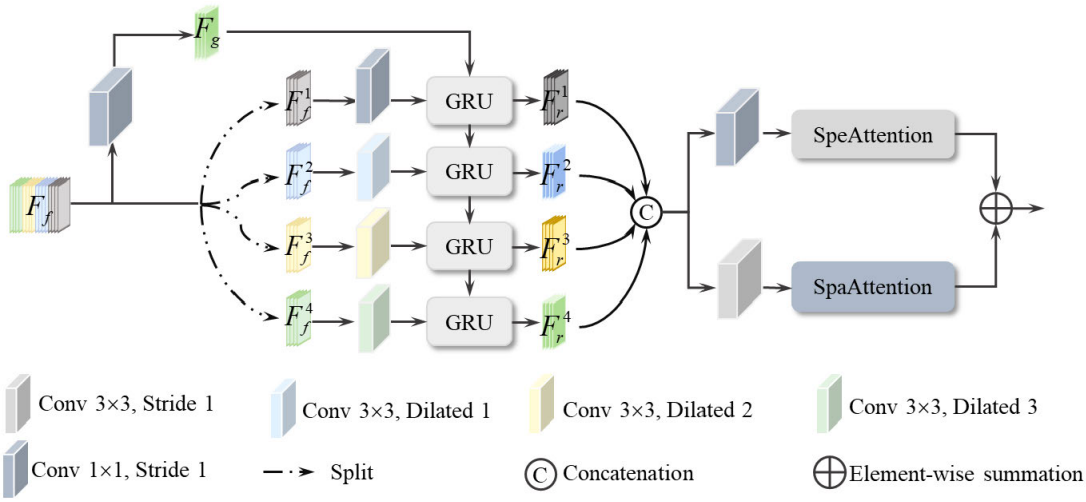


Fig. 5. Framework of the shared reconstruction module. F_f is the fused feature, F_g is the global spectral information, F_r^1, F_r^2, F_r^3, and F_r^4 are the discriminative features learned through GRUs.

is composed of STSM loss, SSR loss, and feature similarity (FS) loss, as illustrated in the following:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SSR}} + \alpha\mathcal{L}_{\text{FS}} + \beta\mathcal{L}_{\text{stsm}} \tag{21}$$

where $\mathcal{L}_{\text{SSR}}$ is the unsupervised loss function for the SSR task, α and β are the tradeoff parameters, $\mathcal{L}_{\text{FS}}$ is feature similarity (FS) loss to allow the two tasks to have similar features before reconstruction, and $\mathcal{L}_{\text{stsm}}$ is the supervised loss function for the STSM task. $\mathcal{L}_{\text{stsm}}$ is illustrated in the following:

$$\mathcal{L}_{\text{stsm}} = 1 - \text{SSIM}(\widetilde{X}, X) + \text{MSE}(\widetilde{X}, X) + \text{SAM}(\widetilde{X}, X) \tag{22}$$

where $\widetilde{X}$ is the reconstructed image, X is the ground truth, SSIM is the structural similarity [56], MSE is the mean square error loss, and SAM is the spectral loss for high spectral fidelity. $\mathcal{L}_{\text{SSR}}$ and $\mathcal{L}_{\text{FS}}$ are illustrated as follows:

$$\mathcal{L}_{\text{SSR}} = \mathcal{L}_{\text{SSR}}^{\text{spectral}} + \mathcal{L}_{\text{SSR}}^{\text{spatial}} \tag{23}$$

$$\mathcal{L}_{\text{SSR}}^{\text{spectral}} = \mathcal{L}_{\text{mse}}^{\text{down}} + \mathcal{L}_{\text{sam}}^{\text{down}} + \mathcal{L}_{\text{sam}} \tag{24}$$

$$\mathcal{L}_{\text{mse}}^{\text{down}} = \text{MSE}(\text{down}(Z), X) \tag{25}$$

$$\mathcal{L}_{\text{sam}}^{\text{down}} = \text{SAM}(\text{down}(Z), X) \tag{26}$$

$$\mathcal{L}_{\text{sam}} = \text{SAM}(\text{GS}(Z), \text{Up}(X)) \tag{27}$$

$$\mathcal{L}_{\text{SSR}}^{\text{spatial}} = \text{MSE}(\text{grad}(Z), \text{grad}(X)) \tag{28}$$

$$\mathcal{L}_{\text{FS}} = \text{MSE}(\text{down}(F_{\text{SSR}}), F_{\text{stsm}}) \tag{29}$$

where down() is the downsample operation with the bicubic interpolation algorithm, GS() is the Gaussian filter operation with the size of 5 × 5 and standard deviation of 0.5 to blur the image, Z is the target high spatial resolution HS image, grad() is the gradient function with aggregating the vertical direction and horizontal direction, and F_{SSR} and F_{STSM} are the feature maps of the SSR network and the STSM network before the reconstruction module, respectively.

III. Experiments

In this section, we conducted experiments on two datasets to evaluate the performance of the proposed USTSFM. In addition, several ablation studies are also conducted to investigate the influence of the shared reconstruction module and loss function.

A. Datasets and Training Details

The first Yellow River Estuary dataset (YRE) consists of two MS–HS image pairs captured by Sentinel-2 and ZY1-02D in YRE, China. The Sentinel-2 MS images were captured on December 8, 2019, and February 11, 2021, respectively, with the size of 5022 × 2074, and four 10-m spatial resolution bands (e.g., red—band 4, green—band 3, blue—band 2, and near-infrared (NIR)—band 8) were selected for our study, The ZY1-02D HS images were captured on December 8, 2019, and February 10, 2021, with the size of 1674 × 758, and 151 effective bands were selected for our study.

The second DongYing dataset (DY) also consists of two MS–HS image pairs captured by Sentinel-2 and ZY1-02D in DY, China. The Sentinel-2 MS images were captured on December 8, 2019, and February 11, 2021, respectively, with the size of 4956 × 5091, and four 10-m spatial resolution bands (e.g., red—band 4, green—band 3, blue—band 2, and NIR—band 8) were selected for our study. The ZY1-02D HS images were captured on December 11, 2019, and February 10, 2021, with the size of 1652 × 1692, and 151 effective bands were selected for our study.

Each dataset is divided into a training set and a testing set; in the training set, the MS images and the corresponding HS images are cropped into patches with the size of 216 × 216 and 72 × 72, respectively. For the testing set, the MS images and the corresponding HS images are cropped into patches with the size of 297 × 297 and 99 × 99, respectively, to more intuitively show the fusion performance. The epoch and batch size are set to 200 and 16, respectively, and the initial learning rate is set to 0.001 and decreases 0.5 in every ten epochs. The proposed USTSFM is equipped with a graphics processing unit (GPU) NVIDIA RTX A6000 with 64 GB, and the compared methods are performed in the MATLAB R2020b framework with 64-GB Intel Core i9-10920X.

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TABLE II
EFFECTIVENESS ANALYSIS OF THE NUMBER N ON THE TWO DATASETS

Dataset	N	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR	Para(M)
YRE	1	3.9034	4.2829	0.8039	0.0256	0.9245	22.9811	**3.2046**
	2	**3.7720**	**4.1660**	**0.8263**	**0.0244**	**0.9253**	**23.5428**	5.6787
	3	4.4103	4.4030	0.8122	0.0267	0.9156	22.7593	8.1527
	4	6.1513	5.0225	0.8112	0.0259	0.9005	23.2688	10.6268
	5	4.3224	4.3282	0.8094	0.0261	0.9205	22.8829	13.1009
DY	1	5.2762	6.0806	**0.8571**	0.0274	0.9134	21.9631	**3.2046**
	2	5.2749	**5.4497**	0.8564	**0.0266**	**0.9251**	**22.6966**	5.6787
	3	**5.2691**	6.2426	0.8461	0.0264	0.9182	22.0893	8.1527
	4	5.8366	7.4789	0.8398	0.0275	0.8959	21.6697	10.6268
	5	5.4605	7.0169	0.8303	0.0289	0.8820	21.6981	13.1009

B. Compared Methods and Evaluation Indices

Because most currently available integrated spatio-temporal–spectral methods are used on higher temporal resolution HS images, they are inapplicable to spatio-temporal–spectral tasks of low spatial, temporal resolution HS images with high spatial, temporal resolution MS images. To verify the superiority of the USTSFM, three state-of-the-art SSR methods (e.g., joint sparse and low-rank learning (J-Slol) [57], Arad [58], Aradplus [59], spatial-spectral residual attention network (SSRAN) [60]) and a newly proposed spatio-temporal–spectral fusion method (e.g., GLMM-STSF [53]) are selected for comparison.

To evaluate the fusion performance quantitatively, six commonly used image quality indexes, including spectral angle mapper (SAM) [61], error relative global dimensionless in synthesis (ERGAS) [62], SSIM, root-mean-square error (RMSE), universal image quality index (UIQI) [63], and peak signal-to-noise ratio (PSNR). Higher SSIM, UIQI, and PSNR and lower SAM, ERGAS, and RMSE indicate better fusion performance.

C. Ablation Studies

1) Effectiveness of the Number of SSICL: We conduct experiments on the two datasets to investigate the effectiveness of the number of SSICL on USTSFM performance. Since the USTSFM is an unsupervised network, we use the fusion results at the reduced resolution as a benchmark to investigate the number of K. Table II illustrates the average results of the testing set; we can see that when $N = 2$, the proposed USTSFM gains the best fusion performance on the YRE dataset and also achieves a competitive performance on the DY dataset, so we set $N = 2$ in the following experiments.

2) Effectiveness of the STFB: To discuss the effectiveness of STFB on USTSFM performance, we conduct experiments at the reduced resolution on the two datasets. Table III illustrates the average metrics on the testing set; we can see that the proposed USTSFM gains the best fusion performance with STFB on the two datasets, which demonstrates that STFB can improve the spatial–spectral fidelity of the model to a certain extent.

3) Effectiveness of Parameters Sharing and Global Spectral Guidance in the Reconstruction: To illustrate the influence of parameter sharing in the reconstruction and guidance with global spectral information, we conduct reduced-resolution experiments on the two datasets, as shown in Tables IV and V.

TABLE III
EFFECTIVENESS ANALYSIS OF STFB ON THE TWO DATASETS

Dataset	Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
YRE	w/o STFB	4.1911	4.4199	0.8127	0.0265	0.9231	22.7319
	USTSFM	**3.7720**	**4.1660**	**0.8263**	**0.0244**	**0.9253**	**23.5428**
DY	w/o STFB	5.6217	5.8743	0.8492	0.0280	0.9206	22.1893
	USTSFM	**5.2749**	**5.4497**	**0.8564**	**0.0266**	**0.9251**	**22.6966**

TABLE IV
EFFECTIVENESS ANALYSIS OF PARAMETERS SHARING AND THE GLOBAL SPECTRAL GUIDANCE IN THE RECONSTRUCTION ON THE YRE DATASET

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR	Para(M)
w/o PS	4.2810	4.6815	0.8079	0.0280	0.9134	22.2754	5.8909
w/o GS	4.3041	4.3745	0.8090	0.0260	0.9249	22.7155	**5.6737**
USTSFM	**3.7720**	**4.1660**	**0.8263**	**0.0244**	**0.9253**	**23.5428**	5.6787

TABLE V
EFFECTIVENESS ANALYSIS OF PARAMETERS SHARING AND THE GLOBAL SPECTRAL GUIDANCE IN THE RECONSTRUCTION ON THE DY DATASET

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR	Para(M)
w/o PS	5.4761	5.7933	0.8415	0.0284	0.9117	21.8316	5.8909
w/o GS	5.7891	5.5624	0.8391	0.0272	0.9198	22.0159	**5.6737**
USTSFM	**5.2749**	**5.4497**	**0.8564**	**0.0266**	**0.9251**	**22.6966**	5.6787

In these tables, the USTSFM without parameter sharing in the reconstruction and the USTSFM without global spectral information are termed as w/o PS and w/o GS, respectively. We can see that the shared reconstruction module can not only improve the fusion performance but also reduce the model parameters, and the USTSFM without global spectral information has relatively fewer parameters than the proposed USTSFM due to the removal of a convolutional layer. However, the USTSFM with the guidance of global spectral information improves the fusion performance on the five quality indexes. The results indicate that parameter sharing and the guidance of global spectral information can boost the ability to describe spatial–spectral information.

4) Effectiveness of the Loss Function: To explore the influence of the loss function on USTSFM performance, we report the performance of the proposed USTSFM by discarding different parts in the total loss function in the reduced-resolution experiments on the two datasets. Specifically, to evaluate the effect of spectral loss $\mathcal{L}_{\text{SSR}}^{\text{spectral}}$ and spatial loss $\mathcal{L}_{\text{SSR}}^{\text{spatial}}$ on the fusion performance, we discard them, termed w/o spectral loss, and w/o spatial loss, respectively. To evaluate the effect of $\mathcal{L}_{\text{SSR}}$, $\mathcal{L}_{\text{stsm}}$, and $\mathcal{L}_{\text{FS}}$ on the fusion performance, we discard them, termed w/o SSR loss, w/o STSM loss, and w/o FS loss, respectively.

The qualitative results are illustrated in Figs. 6 and 7; we can observe that the USTSFM w/o spatial loss and w/o SSR loss have blurring results, which are more similar to the upsampled HS image. Although the USTSFM w/o spectral loss, w/o STSM loss, and w/o FS loss can maintain spatial information, they have more spectral distortion than our proposed with the total loss. The quantitative results are listed in Tables VI and VII, and we can see that the USTSFM w/o spatial loss and w/o SSR loss achieve higher

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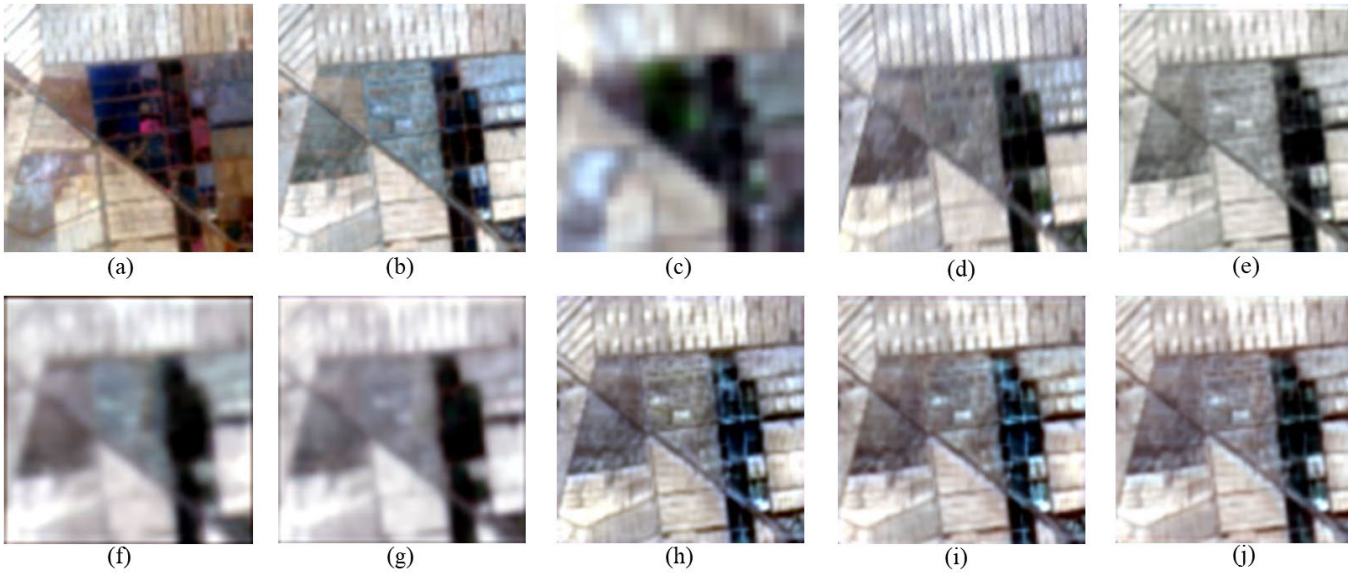


Fig. 6. Fusion results on February 10, 2021 in the YRE dataset. (a) MS image before the target date. (b) MS image at the target date. (c) Upsampled HS image before the target date. (d) Ground truth. The fusion result w/o (e) spectral loss, (f) spatial loss, (g) SSR loss, (h) STSM loss, and (i) FS loss. (j) Fusion result.

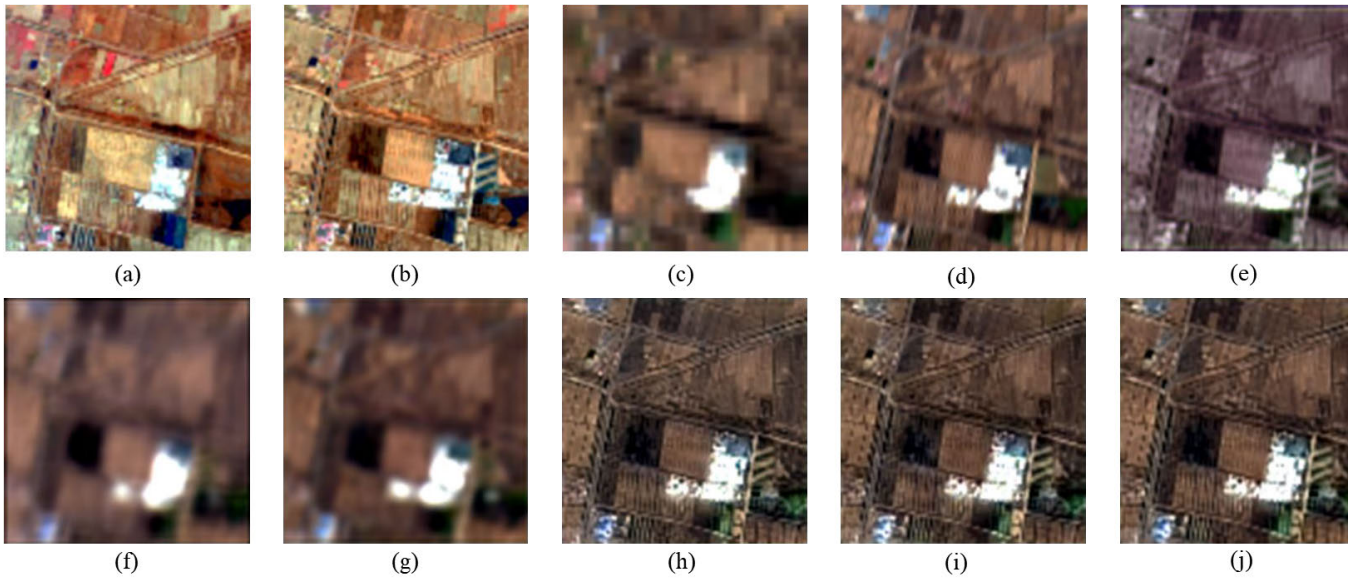


Fig. 7. Fusion results on February 10, 2021 in the DY dataset. (a) MS image before the target date. (b) MS image at the target date. (c) Upsampled HS image before the target date. (d) Ground truth. The fusion result w/o (e) spectral loss, (f) spatial loss, (g) SSR loss, (h) STSM loss, and (i) FS loss. (j) Fusion result.

quantitative indexes than the USTSFM with total loss, which is different from the conclusion of the visual results. We also list the quantitative indexes between the fusion results with the upsampled HS image in Tables VIII and IX; we can see that the fusion results are more similar to the upsampled HS image, which is consistent with the conclusion of the visual results. Based on the above results, we choose the total loss as our loss function to comprehensively consider the degree of preservation of spatial and spectral information.

In addition, we also conduct experiments to illustrate the influence of weights in the loss function (i.e., α and β). Since the purpose of $\mathcal{L}_{\text{FS}}$ and $\mathcal{L}_{\text{stsm}}$ is to make the supervised task (i.e., STSM task) guide the unsupervised task (i.e., SSR task) to have better spatial–spectral fidelity performance. Therefore, we set the same weights (i.e., α and β) for $\mathcal{L}_{\text{FS}}$ and $\mathcal{L}_{\text{stsm}}$ due to the consistent effect. The proposed USTSFM is trained with different values of α and β, and the quantitative results are illustrated in Tables X and XI. As can be seen, the proposed USTSFM achieve the best fusion performance when α and β are set to 1. Therefore, we set α and β to 1 in our proposed USTSFM.

D. Computational Efficiency

We show the test time of different methods at reduced resolution and full resolution in Table XII. It is worth noting

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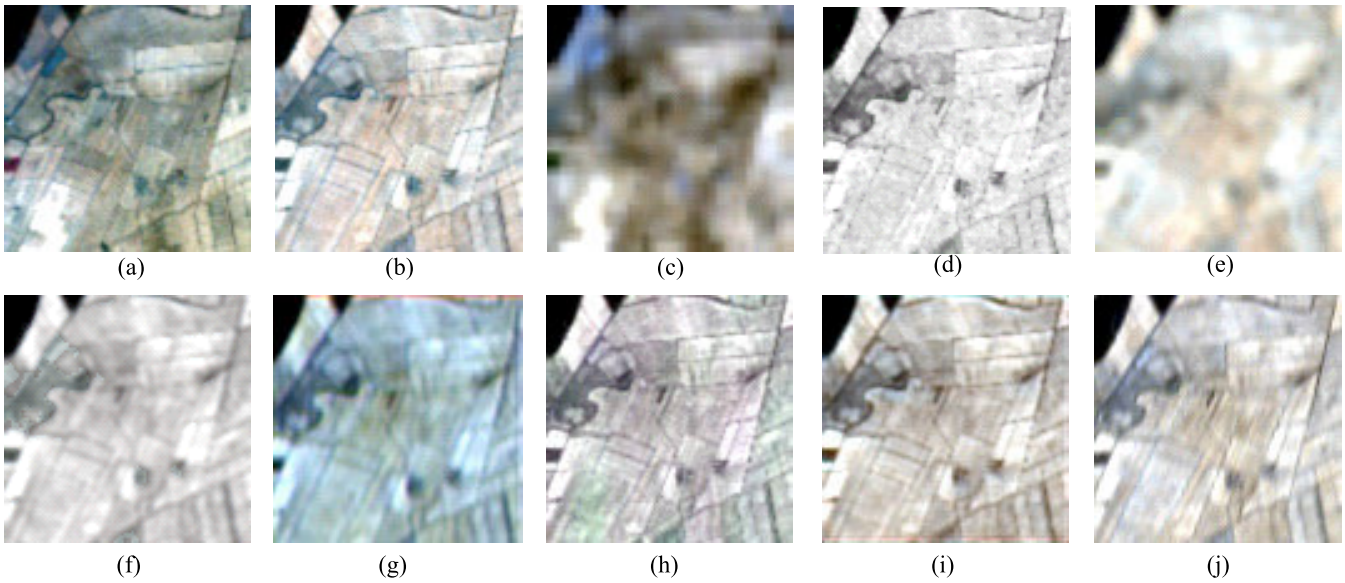


Fig. 8. Fusion results on the YRE dataset in the reduced-resolution experiment. (a) MS image before the target date. (b) MS image at the target date. (c) Upsampled HS image before the target date. (d) J-Slol. (e) Arad. (f) Aradplus. (g) SSRAN. (h) GLMM-STSF. (i) USTSFM. (j) Ground truth.

TABLE VI
EFFECTIVENESS ANALYSIS OF LOSS FUNCTIONS ON THE YRE DATASET

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
w/o spectral loss	5.0842	5.2322	0.7883	0.0310	0.8980	21.5922
w/o spatial loss	**3.5724**	3.6988	0.8515	0.0222	0.9418	24.3398
w/o SSR loss	3.7513	**3.4814**	**0.8785**	**0.0207**	**0.9513**	**25.1919**
w/o STSM loss	3.8285	4.4112	0.7954	0.0262	0.9155	22.7465
w/o FS loss	3.8794	4.3890	0.7968	0.0261	0.9187	22.7920
USTSFM	3.7720	4.1660	0.8263	0.0244	0.9253	23.5428

TABLE VII
EFFECTIVENESS ANALYSIS OF LOSS FUNCTIONS ON THE DY DATASET

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
w/o spectral loss	7.1001	7.8785	0.8297	0.0303	0.8617	19.8744
w/o spatial loss	**4.7971**	5.2681	0.8864	0.0239	0.9461	25.0998
w/o SSR loss	5.1813	**4.6872**	**0.8907**	**0.0221**	**0.9504**	**25.6648**
w/o STSM loss	5.6489	5.9214	0.8444	0.0278	0.9130	21.9492
w/o FS loss	5.2927	7.8889	0.8408	0.0274	0.9183	22.0171
USTSFM	5.2749	5.4497	0.8564	0.0266	0.9251	22.6966

TABLE VIII
QUANTITATIVE EVALUATION RESULTS ON THE YRE DATASET BETWEEN THE FUSION RESULT WITH THE TARGET UPSAMPLED HS IMAGE

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
w/o SSR loss	2.8591	2.5005	0.9627	0.0144	0.9752	27.1906
w/o spatial loss	2.2255	2.2513	0.9753	0.0131	0.9803	27.3153

TABLE IX
QUANTITATIVE EVALUATION RESULTS ON THE DY DATASET BETWEEN THE FUSION RESULT WITH THE TARGET UPSAMPLED HS IMAGE

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
w/o SSR loss	4.0319	3.9651	0.9249	0.0201	0.9652	26.4762
w/o spatial loss	3.7098	3.6471	0.9342	0.0184	0.9701	26.9913

that the reduced-resolution experiment aims to obtain an image with a spatial size of 99 × 99 × 151, and the full-resolution experiment aims to obtain an image with a spatial size of 297 × 297 × 151. We can see that the proposed USTSFM obtains the fusion image with costing the shortest test time, which demonstrates the efficiency of our proposed USTSFM.

TABLE X
QUANTITATIVE EVALUATION RESULTS ON THE YRE DATASET

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
α=β=0.05	3.8694	4.3908	0.7972	0.0262	0.9176	22.7532
α=β=0.2	**3.7572**	4.3657	0.7972	0.0261	0.9181	22.7982
α=β=0.4	4.0747	4.3399	0.8068	0.0257	0.9226	22.9285
α=β=0.6	4.3310	4.2135	0.8198	0.0252	0.9253	23.1817
α=β=0.8	4.2518	4.2674	0.8164	0.0254	**0.9288**	23.0763
α=β=0.9	4.0961	**4.1583**	0.8173	0.0247	0.9285	23.4224
α=β=1	3.7720	4.1660	**0.8263**	**0.0244**	0.9253	**23.5428**

TABLE XI
QUANTITATIVE EVALUATION RESULTS ON THE DY DATASET

Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
α=β=0.05	5.4631	5.5844	0.8482	0.0284	0.9194	22.0357
α=β=0.2	5.7741	5.6217	0.8470	0.0276	0.9192	22.1094
α=β=0.4	**5.2694**	5.6890	0.8531	0.0278	0.9206	22.2240
α=β=0.6	5.6318	5.6062	0.8560	0.0280	0.9217	22.1908
α=β=0.8	5.8219	5.5733	0.8509	0.0271	0.9211	22.3769
α=β=0.9	5.5367	5.5911	0.8549	**0.0264**	0.9227	22.4682
α=β=1	5.2749	**5.4497**	**0.8564**	0.0266	**0.9251**	**22.6966**

E. Experimental Results and Analysis

In this section, we evaluate the proposed USTSFM in reduced resolution and full resolution, respectively. For the reduced-resolution experiments, the degraded test pairs are used for fusion, and the original HS images are used as reference. For the full-resolution experiments, the original test pairs are used for fusion, and the degraded fusion images are compared with the original HS images to evaluate the spectral performance.

1) Reduced-Resolution Experiments: Figs. 8 and 9 display the input images and the fusion results at reduced resolution on

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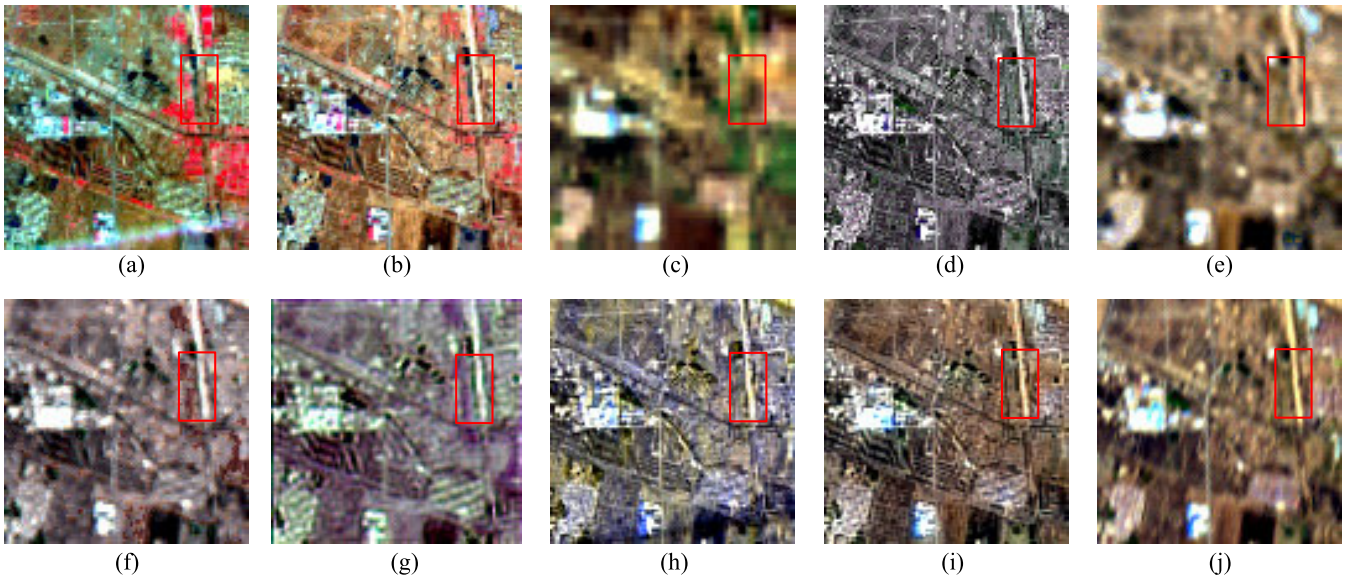


Fig. 9. Fusion results on the DY dataset in the reduced-resolution experiment. (a) MS image before the target date. (b) MS image at the target date. (c) Upsampled HS image before the target date. (d) J-Slol. (e) Arad. (f) Aradplus. (g) SSRAN. (h) GLMM-STSF. (i) USTSFM. (j) Ground truth.

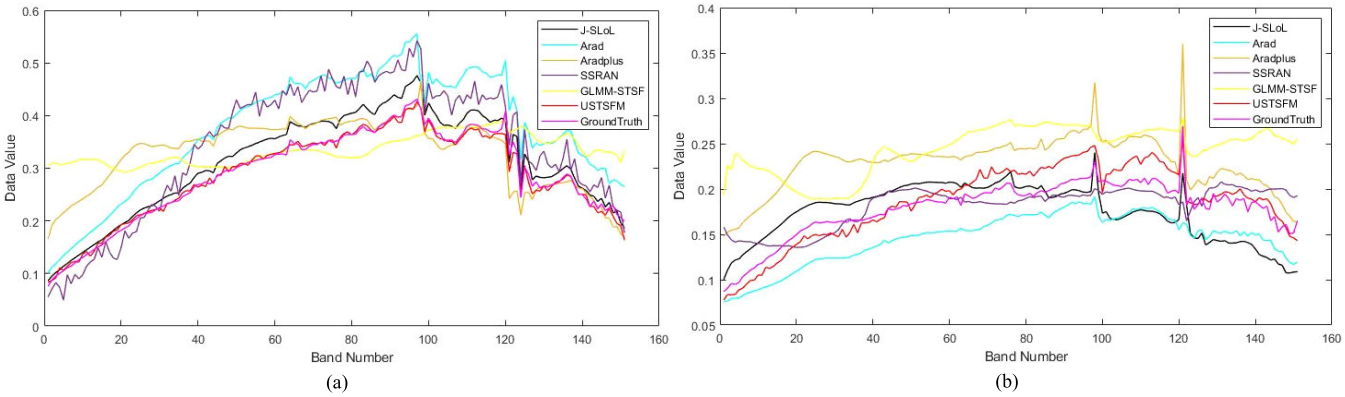


Fig. 10. Spectral profiles of the original HS observations and different fusion results at randomly selected locations on the YRE and DY dataset, respectively. (a) YRE dataset. (b) DY dataset.

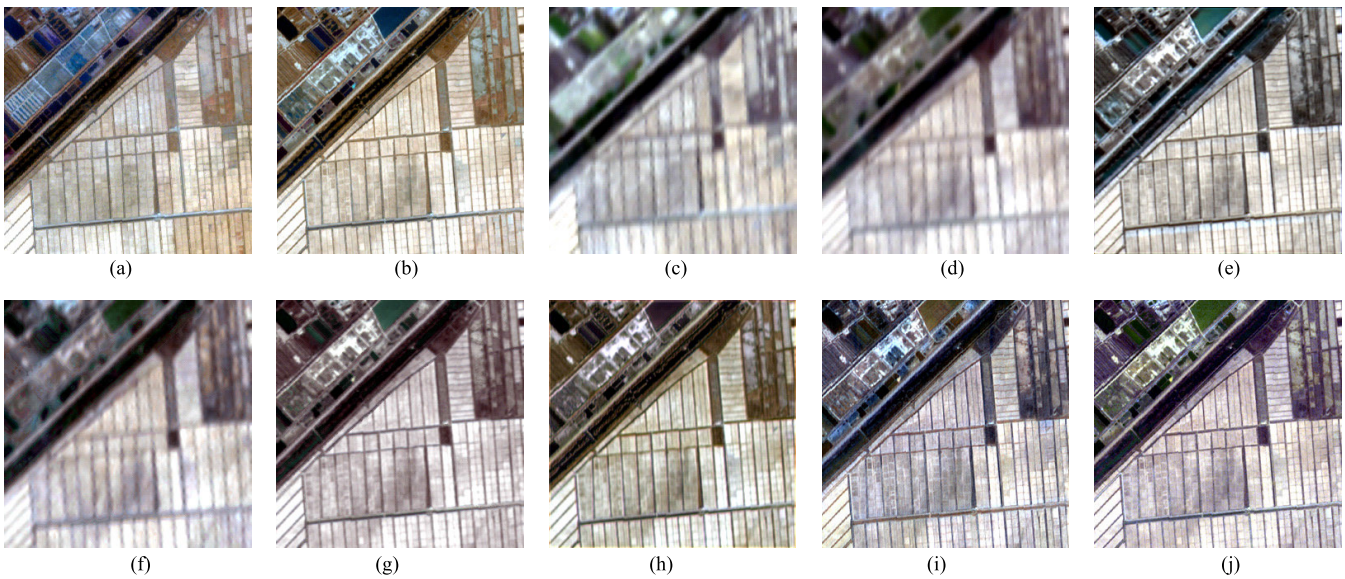


Fig. 11. Fusion results on the YRE dataset in the full-resolution experiment. (a) MS image before the target date. (b) MS image at the target date. (c) Upsampled HS image before the target date. (d) Upsampled HS image before the target date. (e) J-Slol. (f) Arad. (g) Aradplus. (h) SSRAN. (i) GLMM-STSF. (j) USTSFM.

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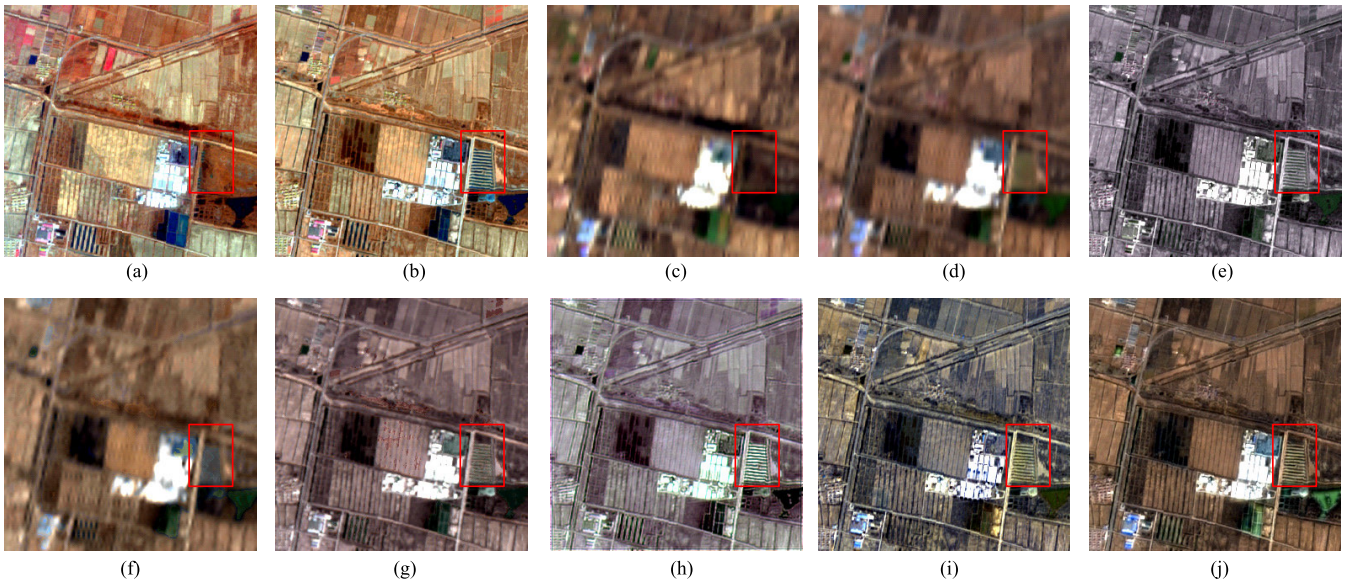


Fig. 12. Fusion results on the DY dataset in the full-resolution experiment. (a) MS image before the target date. (b) MS image at the target date. (c) Upsampled HS image before the target date. (d) Upsampled HS image before the target date. (e) J-Slol. (f) Arad. (g) Aradplus. (h) SSRAN. (i) GLMM-STSF. (j) USTSFM.

the YRE and DY datasets, respectively. From the two figures, we can see that the J-Slol can recover spatial details in the MS image to some extent but it causes obvious spectral distortion in the fusion result. The Arad can well preserve spectral information while serious blurry effects exist. The results of Aradplus and SSRAN have similar poor visual performance, which has obvious spectral distortion and spatial blurring. The results of GLMM-STSF preserve good spatial details but have serious spectral distortion. The results obtained by our proposed USTSFM look more similar to the ground truth. To demonstrate the quality of the fusion images over the entire spectral band, we randomly select a location and illustrate the spectral vectors reconstructed by the comparison methods and our proposed USTSFM, which is shown in Fig. 10. We can see that the spectral vector of our proposed USTSFM is more similar to the ground truth, which demonstrated that our proposed USTSFM can well preserve the spectral information.

To give a quantitative evaluation, we list the average values of the quantitative indexes (e.g., SAM, ERGAS, SSIM, RMSE, UIQI, and PSNR) between the fusion images and the ground truth in Table XIII. Compared with them, the proposed USTSFM obtains the most competitive fusion results on the two datasets. Specifically, the proposed USTSFM achieves higher SSIM, UIQI, and PSNR and lower SAM, ERGAS, and RMSE scores on the two datasets. Generally, from the qualitative and quantitative quality evaluation results, the proposed USTSFM can obtain the fusion result by complementary integrating the spatial and spectral advantages of the MS and HS images.

2) Full-Resolution Experiments: Figs. 11 and 12 display the input images and the full-resolution fusion results. Similar to the results in the reduced-resolution experiments, the results of the comparison methods (e.g., J-Slol, Arad, Aradplus, SSRAN, and GLMM-STSF) have varying degrees of spatial blurring or spectral distortion, while our proposed USTSFM can preserve the spectral information in the HS image and the spatial details in the MS image, but also can predict the land cover changes accurately. To demonstrate the spectral fidelity over the entire spectral band, we downsampled the fusion images and randomly selected a location to plot the spectral vectors, as shown in Fig. 13. The results of the two datasets demonstrated that our proposed USTSFM has a stronger ability in spectral fidelity.

To give an objective evaluation, we downsampled the fusion results at full resolution to achieve a reference-based quality

TABLE XII

TEST TIME COMPARISON AT REDUCED RESOLUTION AND FULL RESOLUTION

Model	Reduced-resolution		Full-resolution	
	CPU(s)	GPU(s)	CPU(s)	GPU(s)
J-SLoL	0.21	-	2.15	-
Arad	2.07	-	20.77	-
Aradplus	1.64	-	16.03	-
SSRAN	0.27	0.03	2.87	0.43
GLMM-STSF	1.94	-	18.29	-
USTSFM	**0.17**	**0.02**	**1.62**	**0.32**

TABLE XIII

QUANTITATIVE EVALUATION RESULTS IN THE REDUCED-RESOLUTION EXPERIMENTS

Dataset	Methods	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
YRE	J-SLoL	8.9231	6.4948	0.7726	0.0336	0.8343	22.2743
	Arad	11.7365	14.0365	0.6920	0.0669	0.6905	14.2569
	Aradplus	5.2799	5.4895	0.7641	0.0323	0.8747	21.5834
	SSRAN	10.0634	12.3682	0.6651	0.0924	0.6726	16.2243
	GLMM-STSF	9.5094	11.6982	0.7692	0.0708	0.7081	15.3341
	USTSFM	**3.7720**	**4.1660**	**0.8263**	**0.0244**	**0.9253**	**23.5428**
DY	J-SLoL	8.9431	8.1134	0.8004	0.0396	0.7814	19.5827
	Arad	11.8096	10.3261	0.6199	0.0987	0.8019	17.8096
	Aradplus	9.8765	14.8922	0.7872	0.0595	0.8005	15.6466
	SSRAN	11.5924	12.0967	0.7461	0.0806	0.7734	18.2246
	GLMM-STSF	10.3265	12.0895	0.6892	0.0864	0.8104	16.3271
	USTSFM	**5.2749**	**5.4497**	**0.8564**	**0.0266**	**0.9251**	**22.6966**

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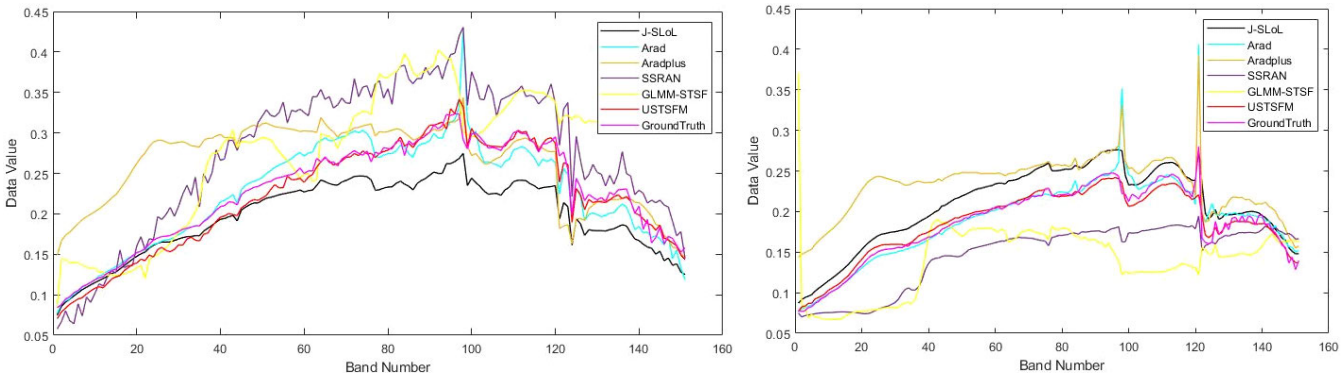


Fig. 13. Spectral signatures at full resolution with the randomly selected location. (a) YRE dataset. (b) DY dataset.

TABLE XIV
QUANTITATIVE EVALUATION RESULTS AT THE FULL-RESOLUTION EXPERIMENTS

Dataset	Model	SAM	ERGAS	SSIM	RMSE	UIQI	PSNR
YRE	J-SLoL	6.3452	7.2620	0.6663	0.0459	0.8029	19.3742
	Arad	6.5605	4.2249	0.8605	0.0240	0.9268	**26.6793**
	Aradplus	12.5264	17.1303	0.4957	0.0911	0.5009	12.1353
	SSRAN	9.8219	11.8642	0.7024	0.0791	0.6018	18.2467
	GLMM-STSF	13.5681	15.2486	0.5634	0.0672	0.5619	14.3591
	USTSFM	**5.2140**	**4.0329**	**0.9171**	**0.0222**	**0.9395**	24.9351
DY	J-SLoL	8.8714	10.8449	0.6182	0.0572	0.7404	18.2520
	Arad	8.3754	5.2294	**0.8980**	0.0216	0.9181	**27.1705**
	Aradplus	10.0837	15.7989	0.7030	0.0663	0.7596	14.9442
	SSRAN	10.3846	9.9428	0.7247	0.0792	0.7483	15.2273
	GLMM-STSF	11.9561	13.8249	0.7605	0.0724	0.7951	15.3873
	USTSFM	**3.0889**	**3.7962**	0.8685	**0.0208**	**0.9586**	25.7701

assessment, and the average values of the quality indexes (e.g., SAM, ERGAS, SSIM, RMSE, UIQI, and PSNR) are listed in Table XIV. We can see that the proposed USTSFM also obtains competitive quality scores. Although Arad obtains higher scores in SSIM and PSNR, the visual results in Figs. 11 and 12 show that it cannot preserve the spatial details in the MS image, and the results are more similar to the HS image. On the whole, the proposed USTSFM can absorb spatial and spectral information from the source images, which can obtain the most competitive fusion results.

IV. CONCLUSION

In this article, we proposed an efficient and effective USTSFM with dual-task interactive learning for remote sensing images. In USTSFM, we proposed two collaborative tasks (i.e., the STSM and SSR) to complementary represent the relationship among spatial, temporal, and spectral. Specifically, we designed a spatial–spectral interactive compensation learning to make the two tasks benefit from each other. In addition, we meticulously designed a reconstruction module with shared weights to simultaneously reconstruct the target images of both the tasks because the SSR task has no corresponding reference HS image (label). Moreover, the proposed USTSFM is jointly optimized through supervised and unsupervised learning. The qualitative and quantitative results demonstrated that the USTSFM not only can absorb the spatial and spectral information from the source images but also can predict the land cover changes accurately.

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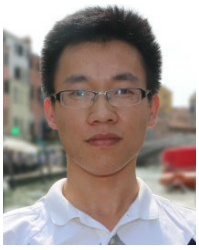
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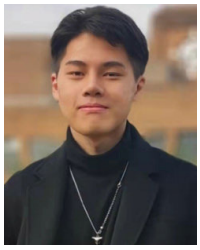
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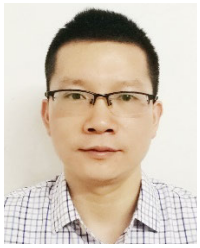
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